QIP - Quantum Information Processing with the IBM Quantum Experience

Signature Sheet

Student's Name _____ Partner's Name _____

Before the Lab See main text. **Pre-Lab Discussion Questions**

It is your responsibility to discuss this lab with an instructor before your first day of your scheduled lab period. This signed sheet must be included as the first page of your report. Without it you will lose grade points. You should be prepared to discuss at least the following before you come to lab:

1. What technology is used, what is a qubit, discuss at least three different methods to encode a qubit, what is the promise of quantum computing. How are the qubits coupled to each other ?
2. What is the electrical circuit used to create a Josephson junction qubit ? What is the energy eigenspectrum of the circuit ? Which levels constitute the qubit ?
3. How is a qubit controlled: develop intuition from a spin-1/2 system with magnetic moment in a magnetic field, how can the qubit be rotated, how can we measure the expectation value of the qubit along various axes ?
4. What is the most general state of a qubit ? What is the most general unitary transformation ?
5. Optional: full derivation of how a qubit is controlled with microwave radiation
6. What is entanglement ? How can we use a controlled-NOT operation and single-qubit control to create the states $(|00\rangle \pm |11\rangle)/\sqrt{2}$, $(|01\rangle \pm |10\rangle)/\sqrt{2}$?

Staff Signature _____ Date _____

Completed before the first day of lab? (Circle one) Yes / No

Mid-Lab Discussion Questions

By day 3 or 4 of this lab, you should have successfully ???

1. Record and report your half-wave plate calibrations.
2. Create a graph that shows the purity of your Bell state and show it to a GSI for a signature. Also, show your quantified bell state (coefficients C_1,C_2, and phase ϕ).
3. Which pairs of half-wave plate angles will you use for your measurement of S? Why are these the best values to use for your entangled state? Make sure you have math to back you up.
4. Please measure the output power of the laser diode after the optical Isolator and show and tell the professor or GSI. Document this number.

Staff Signature _____ Date _____

Completed by day 4 of lab? (Circle one) Yes / No

Checkpoint Signatures

1. Violet Beam Path

   Staff Signature _____

2. Angle

   Staff Signature _____

3. Infrared Beam Path

   Staff Signature _____

4. Arbitrary Offset

   Staff Signature _____

5. Bell-State Characterization

   Staff Signature _____